



Foreword

Thank you for purchasing this asset! I truly appreciate your support, and I hope it helps bring your tree creations to life. This tool was designed with flexibility and ease of use in mind, allowing you to generate unique trees with adjustable parameters for trunks, branches, and leaves.

Whether you're creating dense forests, mystical landscapes, or standalone hero trees, this system provides a solid foundation while still giving you full control over the details.

If you have any questions, feedback, or suggestions, feel free to reach out—I'd love to hear from you and help in any way I can.

Additionally, I invite you to check out my other assets available in the Unity Asset Store.

I offer a variety of tools and utilities designed to enhance your development process and bring your creative visions to life.

Enjoy creating, and happy developing!

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1. Introduction

MysticForge: Procedural Tree Generator

Creating unique and realistic trees for your Unity project has never been easier!

MysticForge: Procedural Tree Generator is a powerful and flexible tool that allows you to generate an unlimited variety of trees, from towering pines to elegant willows and everything in between.

Unlike static tree assets, this tool provides complete **procedural control**, ensuring that no two trees look the same. Users can customize **trunks, branches, branchlets, and**

leaves through an intuitive **Inspector-based editor**, allowing for endless variations. Additionally, real-time statistics such as **triangle, vertex, and edge count** provide insights into performance, ensuring that your trees remain optimized for any game environment.

Enhanced Workflow & Usability

- **Integrated Material Editing** – Adjust material properties for both bark and leaves directly within the editor—no need to switch to the Inspector.
- **Quick Material & Model Selection** – Easily swap materials and models via dropdown menus, removing the hassle of dragging and dropping assets.
- **6 Custom Presets** – Jumpstart your workflow with pre-made tree setups, making it easier to understand how different parameters affect tree generation.

Key Features

- ✓ **Procedural tree generation** with unlimited customization
- ✓ Adjustable **trunk, branches, branchlets, and leaves**
- ✓ **Live scene visualization**
- ✓ Custom **bark and leaf textures & models**
- ✓ **Direct material editing** inside the editor
- ✓ **Quick material & model selection via dropdowns**
- ✓ **6 pre-made presets** for an easy starting point
- ✓ Real-time **triangle, vertex, and edge count display**

2. Installation & Setup

Requirements

Before using **MysticForge: Procedural Tree Generator**, ensure that your Unity project meets the following requirements:

- **Unity Version:** 2021.3.26f or newer

- **Compatible Render Pipelines:**
 -  **Built-in Render Pipeline**
 -  **Universal Render Pipeline (URP)**
 -  **High Definition Render Pipeline (HDRP)** (Material compatibility may require minor adjustments)

Installation

1. Purchase and download **MysticForge: Procedural Tree Generator** from the Unity Asset Store.
2. In Unity, open the **Package Manager** (*Window → Package Manager*), locate the asset, and click **Import**.
3. Once imported, the tool is ready to use—no additional setup required.

Accessing the Tool

To open the tree generator:

1. Navigate to **Tools** in Unity's top menu bar.
2. Click **MysticForge: Procedural Tree Generator**.
3. The editor window will open, allowing you to start customizing and generating trees immediately.

Getting Started

- **Preset-Based Workflow** – The tool includes **6 pre-made presets** to help you quickly understand how tree generation works. Simply select a preset and modify it to your needs.
- **No Additional Setup Required** – Since the tool runs within the editor, there's no need to attach scripts or manually configure components—just open the window and start generating trees!

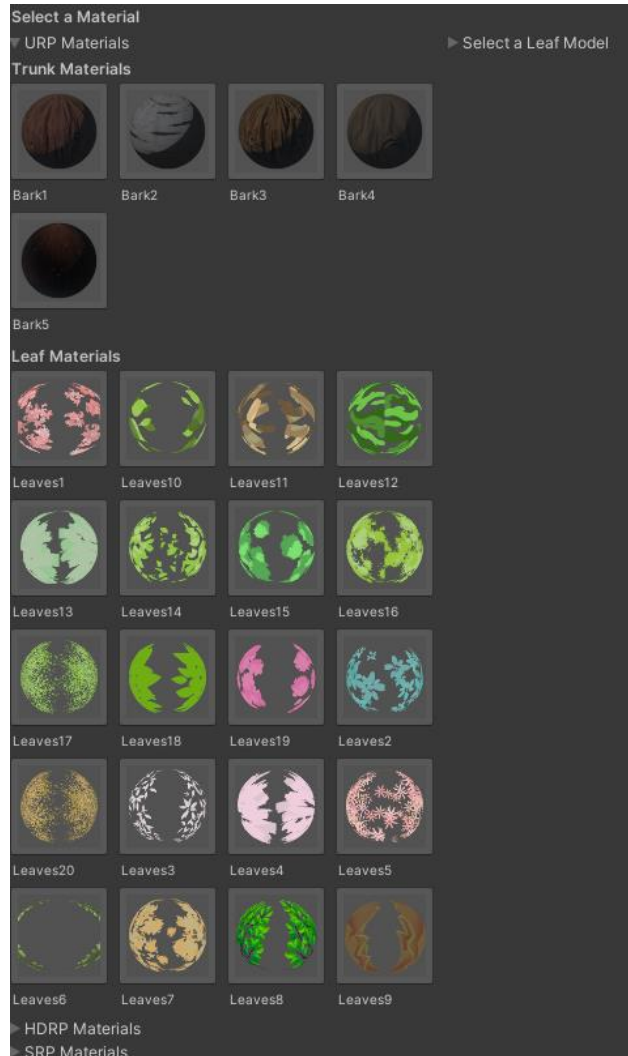
3. Getting Started

Creating a Tree: Step-by-Step Guide

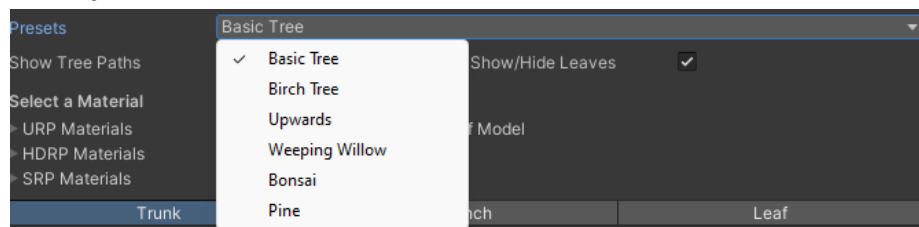
Follow these simple steps to generate your first tree with **MysticForge: Procedural Tree Generator**:

1. **Open the Tool**

- a. Go to **Tools → MysticForge: Procedural Tree Generator** in the Unity top menu bar to open the editor window.
2. **Select a Preset** (optional)
 - a. Select a bark and leaf material and a leaf model from the dropdowns depending on your chosen RP:

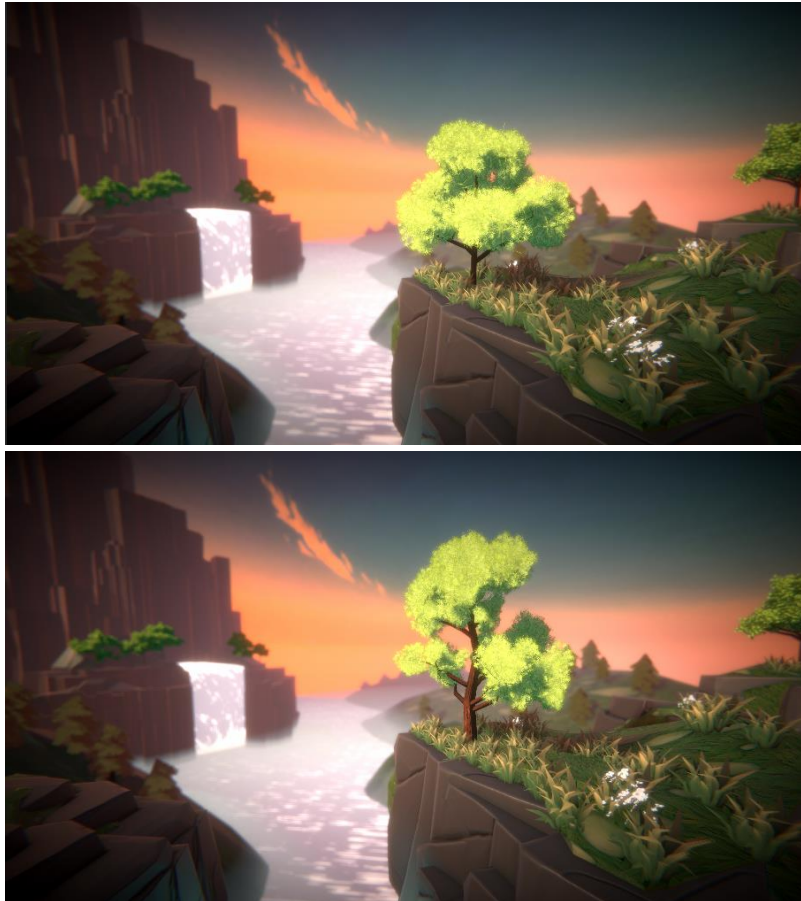


- b. If you're new to the tool, start by selecting one of the 6 **pre-made presets** available in the editor. This will provide you with a base tree setup to modify.



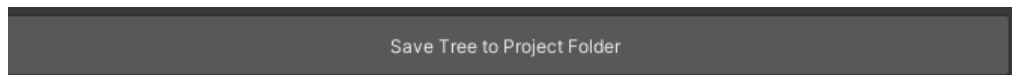
3. Adjust Parameters

- a. Begin customizing the tree by adjusting key parameters such as **tree height, trunk radius, and branch settings**. You'll see the tree update live in the **Scene View** as you make changes.



4. Export the Tree

- a. Once you're satisfied with the tree's appearance, click **Export** to save the tree's mesh and corresponding prefab. Button is at the bottom of the tool:



Basic Parameters: Overview of Essential Settings

Here are some of the **core parameters** you'll use to shape your tree:

Trunk settings:

- **Trunk Height**
 - Adjust the height of your tree by sliding the **Tree Height** slider. This changes the vertical size of the entire tree from the base to the top of the canopy.
- **Trunk Radius**
 - Control the thickness of the tree's trunk with the **Trunk Radius** slider. This affects how wide or thin the trunk appears. You can also tweak the **Trunk Radius Curvature** to make the trunk more tapered or bulbous.
- **Trunk Bending**
 - Change the strength of the trunk bending for a more natural looking tree.

Branch settings:

- Use the **Branch Count** slider to control how many branches sprout from the trunk.
- The **Branch Bending** slider lets you adjust the angle and curvature of branches, creating a more natural, organic look.
- **Branch Radius**
 - Control the thickness of the branch's with the **Branch Radius** slider. This affects how wide or thin the branch appears. You can also tweak the **Branch Radius Curvature** to make the branch more tapered or bulbous.
- **Branch length**
 - Control the length of the branch's with the **Branch Length** slider. This affects how long the branch will be.

Branchlet settings:

- Use the **Branchlet Count** slider to control how many branches sprout from the branches.
- The **Branchlet Bending** slider lets you adjust the angle and curvature of branchlets, creating a more natural, organic look.
- **Branchlet Radius**
 - Control the thickness of the branchlets with the **Branchlet Radius** slider. This affects how wide or thin the branchlet appears. You can also tweak the **Branchlet Radius Curvature** to make the branch more tapered or bulbous.
- **Branchlet length**
 - Control the length of the branch's with the **Branchlet Length** slider. This affects how long the branchlet will be.

Leaves Settings

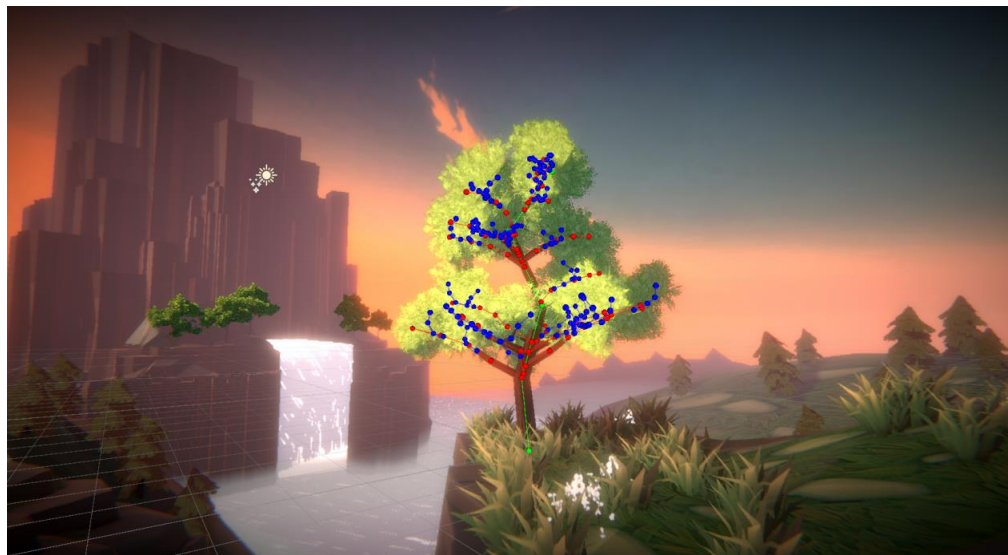
- **Number of leaves**
 - Control the amount of leaves on each branch, branchlet or the trunk itself.
- **Leaf sizing**
 - Control the size of the leaves with the **Leaf Size**, **Leaf Scaling Vector3** and **Leaf Size Variation** parameters
- **Leaf rotation and placement position**
 - Control the rotational direction and position of the leaves

You'll see real-time updates in the **Scene View** as you adjust each parameter.

Live Editing: Real-Time Changes in the Scene View

As you tweak the settings, you'll see the changes reflected instantly in the **Scene View**. This dynamic preview makes it easy to experiment and fine-tune your tree's appearance.

- **Real-Time Updates**
 - Changes made in the **Inspector** (height, radius, branch count, etc.) will immediately affect the tree, allowing you to see how adjustments influence the overall shape and size.
- **Scene Pathing**
 - You can also enable the option to show **trunk, branch and branchlet pathing** directly in the Scene View. This shows how the branches are growing out from the trunk, which helps you visualize the tree's structure in 3D space.



5. Advanced Features

Tree Parameters Overview

Trunk Parameters

- **Trunk Height**

→ The height of the tree trunk, affecting how tall the tree grows from its base.
Higher values result in taller trunks.

- **Trunk Radius**

→ Determines the width of the trunk at the base. Increasing the value will make the trunk thicker.

- **Trunk Radius Curvature**

→ Controls the tapering of the trunk as it moves upwards. A value close to 1 creates a more conical trunk, whereas values closer to 0 make it appear more cylindrical.

- **Trunk Radius Noise**

→ Adds randomness to the trunk's radius, creating a more organic shape with slight variations along its height.

- **Trunk Subdivision**

→ This controls the number of subdivisions along the trunk, affecting the smoothness and polygonal detail. Higher values result in smoother trunks.

- **Trunk Crinkliness**

→ This parameter introduces crinkling or slight deformation/winding to the trunk, which can give the tree a more natural, weathered look.

- **Trunk Segments**

→ Defines how many horizontal segments are used in the trunk, allowing for more flexibility in how the trunk is constructed. Increasing the segments adds more detail but can affect performance.

- **Trunk Bending**

→ Slight bending of the trunk, controlled by this value. A higher value gives the trunk a more curved, dynamic shape.

- **Trunk Material**

→ If you decide to add your own trunk material rather than using one from the dropdown you can add it here.

- **Material properties**

→ Adjust the material settings directly within the editor of the chosen material.

Attention: Only the supplied materials work with this function: If you wish to get the functionality with your own materials, please send me a message.

Branch Parameters

- **Number of Branches**

→ Controls how many branches emerge from the trunk. Increasing this will add more complexity to the tree structure.

- **Branch Height Min & Max**

→ These values determine where along the trunk the branches will appear. The minimum and maximum height set the range in which branches will spawn.

- **Branch Radius**

→ Defines the width of the branches. Higher values give thicker branches, while lower values create thinner branches. **A notice:** Branches maximum radius will only be as thick as the position of their trunk attachment point.

- **Branch Length**

→ Controls how long the branches are. Longer branches stretch outward from the trunk, while shorter ones are more compact.

- **Branch Radius Curvature**

→ Adjusts how much the radius of branches changes as they extend from the trunk. A value of 1 will create branches that get thinner towards the end, while values closer to 0 will make branches appear more uniform in thickness.

- **Branch Radius Noise**

→ Similar to trunk radius noise, this parameter introduces irregularity to branch widths, resulting in a more natural, imperfect look.

- **Branch Subdivision**

→ Controls the number of subdivisions per branch. More subdivisions give finer details, but at the cost of performance.

- **Branch Crinkliness**

→ This adds slight winding around themselves to the branches for a more natural look.

- **Branch Segments**

→ Defines the number of segments used to create each branch. More segments increase detail and curvature options.

- **Branch Bending**

→ Adjusts the overall bend of the branches.

- **Branch Angle**

→ Specifies the angle at which the branches grow outward from the trunk. Negative values will tilt branches downward, and positive values tilt them upward.

- **Branch Gravity**

→ The gravity value affects how much branches sag down over their length. Higher values make branches more droopy, while lower values result in more rigid branches.

- **Shorten by Height**

→ This bool decreases the length of the spawned branches depending on their y-spawn position to create a more natural looking tree.

- **Increase Angle by Height**

→ This bool increases the “Branch Angle” of the branches depending on their y-spawn position. Branches that spawn higher will be more angled to create a more natural looking tree.

Branchlet Parameters

- **Number of Branchlets**

→ Controls the number of smaller branches that grow from the main branches. More branchlets increase the tree's complexity.

- **Branchlet Length & Radius**

→ Similar to branch parameters, these define the size and thickness of the smaller branchlets. **A notice:** Branches maximum radius will only be as thick as the position of their branch attachment point.

- **Branchlet Radius Curvature**

→ Adjusts how much the radius of branchlets changes as they extend from the their branch. A value of 1 will create branchlets that get thinner towards the end, while values closer to 0 will make branchlets appear more uniform in thickness.

- **Branchlet Crinkliness & Bending**

→ Adjusts how much bend and crinkle appear on the branchlets. These settings can make the branchlets look more organic.

- **Branchlet Gravity**

→ Affects how much branchlets sag due to gravity. Increasing this value will make branchlets droop more.

- **Branchlet Forward Angle**

→ Controls the angle at which the branchlets grow forward from the branches.

- **Branchlet Subdivision and Segments**

→ Same functionality as the tree trunk or branch logic.

- **Shorten by Height**

→ This bool decreases the length of the spawned branchlets depending on their y-spawn position to create a more natural looking tree. The higher the branchlet, the shorter it is going to be.

Leaf Parameters

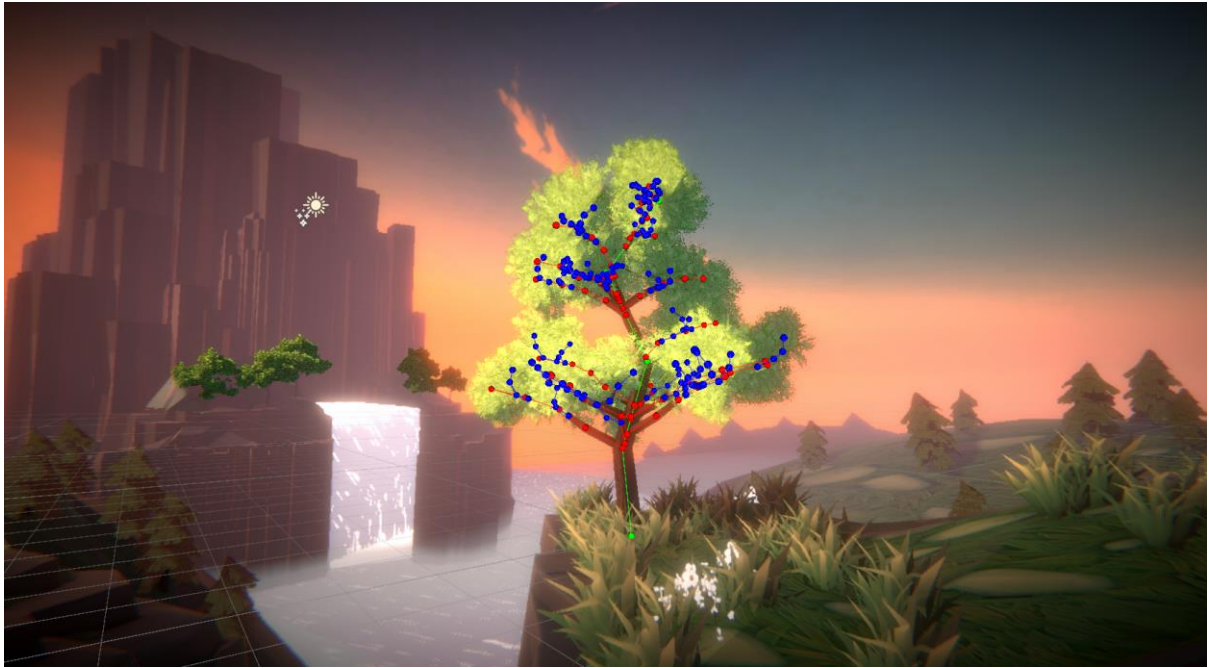
The following parameter logics can be applied to all three different leaf parameters (branch, branchlet and trunk).

- **Number of Leaves**
 - Specifies how many leaves are attached to the tree. Increasing this value results in a fuller tree.
- **Leaf Size**
 - Adjusts the overall size of the leaves. Larger values make the leaves bigger and more prominent.
- **Leaf Scaling Vector3**
 - More precise option to scale the leaves in all three directions.
- **Leaf Size Variation**
 - Adjust this value to randomly scale each leaf with a certain strength to make the leaves more natural looking.
- **Leaf Position Min & Max** determine the range along the branch, branchlet or trunk where leaves will appear. You can use these values to place leaves more densely or sparsely.
- **Leaf Rotation, Forward Rotation & Randomization**
 - These parameters control how the leaves rotate around their position on branches or trunks. The randomization value gives the leaves a more natural, varied look.
- **Leaf Position XYZ & Randomize Leaf Spread**
 - Allows for the fine control of leaf placement. This is useful when you want leaves to align differently or want to adjust their placement position.

Additional Features

Show Tree Paths

The **Show Tree Paths** feature provides a visual representation of the trunk, branches, and branchlets directly in the Scene View. Different colors are used to distinguish each part of the tree, making it easier to analyze the tree's structure.



Show/Hide Leaves

The **Toggle Leaves** function allows users to hide all leaves in the scene, making it easier to focus on the underlying structure of the tree. This is particularly useful when refining trunk and branch configurations.



Material Selection by Render Pipeline

The tool includes an automatic **Material Selection** system based on the user's current Render Pipeline (RP). The available material folders are:

- **HDRP Materials:** Assets/MysticForge ProceduralTreeGenerator/Materials/HDRP/Bark & Assets/Stylized Trees/Materials/HDRP/Leaves
- **URP Materials:** Assets/ MysticForge ProceduralTreeGenerator/ Materials/URP/Bark & Assets/Stylized Trees/Materials/URP/Leaves
- **SRP Materials:** Assets/ MysticForge ProceduralTreeGenerator/ Materials/SRP/Bark & Assets/Stylized Trees/Materials/SRP/Leaves

Users who want to add custom materials for quick selection must place their materials in the respective paths above. The dropdown menus will automatically list any materials placed in these directories, making it easy to switch between different bark and leaf textures. **A notice: You might have to close the generator via Right-Click and “Close Tab” and reopen it from the toolbar to make your changes visible. Dont forget to save your current tree to not lose any progress!**

Leaf Model Selection

Users can select different **Leaf Models** to define the shape and initial structure of leaves. Since the tool uses alpha textures for leaves, different base models significantly impact the final appearance. To add custom leaf models, users should place their .fbx or .obj files in: Assets/ MysticForge ProceduralTreeGenerator/ LeafModels

Once added, these models will appear in the **Select a Leaf Model** dropdown, allowing for quick selection and application. **A notice: You might have to close the generator via Right-Click and “Close Tab” and reopen it from the toolbar to make your changes visible. Dont forget to save your current tree to not lose any progress!**

Mesh Information

The mesh information about vertices, edges, and faces is crucial for game optimization because it directly impacts the performance of rendering and physics calculations. By visualizing these details in your editor window, users can identify high-poly areas in the mesh and make informed decisions about simplifying or optimizing those sections. For instance, reducing the number of vertices or triangles in less critical parts of the model can lower the computational cost, improving the game's performance. The total counts give users an overview of the complexity, and focusing on sections like the trunk, branches, and leaves allows them to prioritize optimization where it matters most—

often in areas that are more visible or heavily interacted with during gameplay. By tracking these values, users can ensure that the meshes are efficient without sacrificing visual quality, enhancing both the user experience and overall game performance

Generated Mesh Info			
Mesh Part	Vertices	Triangles	Edges
Trunk	36	54	89
Branch	612	918	1513
Branchlet	1160	2400	6000
Leaves Branch	2856	1428	3570
Leaves Branchlet	4800	2400	6000
Leaves Trunk	112	56	140
Total	Vertices	Triangles	Edges
Total	9576	7256	17312

Additional Shader: Foliage Toon Shader with Wind Animations

This asset includes a specialized toon shader designed for trees, featuring built-in wind animations. The shader supports both **vertex position animation** (for bending entire leaves or branches) and **UV-based animation** (for subtle texture-based movement). This allows for highly customizable wind effects that can be fine-tuned to match various environmental conditions. Users can easily create their own leaf textures and are encouraged to check the included textures to understand the best practices. If users need assistance, they can contact the developer.

Foliage Toon and Billboard Shader Settings and Their Functions:

This asset includes different shaders. The most important one for the tree generation is the “LeavesShader”. The “BarkShader” works the same way. **A notice: These shaders mainly provide a good working solution in URP. I have included a similar shader for HDRP users but it is not as optimized as the URP shader. If you have any suggestions or improvements, definitely write me a message.**

- **Main Texture** – The primary texture applied to the leaves.
- **Texture Alpha** – Controls the transparency of the leaf texture.
- **Main Color** – Adjusts the overall color of the leaves.
- **Shadow Texture** – An optional texture used to define shadow details.
- **Shadow Color** – The color applied to shadowed areas of the leaves.
- **Shadow Pattern Color** – A secondary shadow color for additional variation.
- **Shadow Step** – Defines the starting position of the shadows.

- **Wind Direction UV** – Determines the wind direction applied to the leaf texture's UV coordinates.
- **Wind Speed UV** – Adjusts the speed at which the leaf texture moves to simulate wind.
- **Wind Noise UV** – Introduces randomization to the wind effect, preventing repetitive motion.
- **Bending Direction Vertex** – Specifies the bending direction for the entire leaf geometry.
- **Bend Strength Vertex** – Controls how strongly the leaves bend in response to wind forces.
- **Wind Speed Vertex** – Determines how fast the leaves bend and sway.
- **Noise Wind Vertex** – Adds variation to vertex-based wind movement to make the animation appear more organic.
- **Billboard Size**– Increases the size of the planes/quads. Billboards automatically face the camera direction. It is advised to use the single quad Leaf Model with the Billboard Foliage Shader

These settings provide a high degree of customization, enabling users to create a variety of tree and foliage animations that respond naturally to wind. If users wish to create custom leaf textures, they should reference the included textures for guidance.

6. Support

For assistance with the Procedural Tree Generator, please use the following contact methods:

Email: Directly reach out to me at info.past12pm@gmail.com. Provide detailed information about your issue or inquiry to help me address your needs effectively.

Website Contact: You can also contact me through my website. Visit the contact page for additional ways to get in touch.

I am dedicated to helping you with any questions or issues you may have and will do my best to provide prompt and helpful responses.